

Approximate Causal Abstractions

- Abstract:

M_L : 2 high dimensional vector valued variables

- $\vec{T}$: sea surface temperatures

- $\vec{W}$: wind speeds

s.t. $\vec{T} \& \vec{W} \in V_L$

• Assumption: wind speed is a cause of sea temperature

- $\vec{T} = f_L(\vec{W}, \vec{U})$ where $\vec{U}$ is noise

- $\vec{U}$ may be a confounder of

$\vec{W} \& \vec{T}$, additional causal

relationship: $\vec{W} = g_L(\vec{U})$

System can be modeled abstractly using: $E = f_H(c, \vec{U})$

• surjective state mapping τ from $R(\vec{T}, \vec{W})$ to $R(E, c)$

- Primitive event is a formula of the form $X=x$, $X \in V \& x \in R(X)$.

- A causal formula is one of the form:

$[Y_1 \leftarrow y_1, \dots, Y_k \leftarrow y_k] \varphi$,

① φ is a boolean combination of primitive events

② $Y_i \in V \& y_i \in R(Y_i)$

③ abbreviated as: $[\vec{Y} \leftarrow \vec{y}] \varphi$

④ $k=0? \rightarrow \varphi$.

φ would hold if Y_i were set to y_i for $i=1, \dots, k$.

- Induced intervention mapping:

- forces high level actions to be logically derived from how the variables were grouped in the first place

- Recall: Restriction set *

$$Rst(V, \vec{x}) = \{ \vec{v} \in R(V) : \vec{x} \text{ is the restriction of } \vec{v} \text{ to } \vec{x} \}$$

- set of all possible ways the whole system could look, given that we've forced certain parts to stay fixed.

- In other words, if you have a system of variables (V) & you manually fix a few of them (x) to values (x), the restriction set * is the cloud of every possible state the entire system could be in while keeping those specific variables fixed.

$$- \tau(Rst(V_L, x)) = Rst(V_H, y) ***$$

- Definition 2.6: (constructive abstraction)

$$\text{If } V_H = \{y_1, \dots, y_n\}$$

$\tau: R_L(V_L) \rightarrow R_H(V_H)$ is constructive

if $E \vec{P} = \{ \vec{z}_1, \dots, \vec{z}_{n+1} \}$ of V_L [$\vec{z}_i$ is a bucket of low level endogenous vars]

mappings $\tau_i: R(\vec{z}_i) \rightarrow R(y_i)$ for $i=1, \dots, n$

s.t $\tau = (\tau_1, \dots, \tau_n)$ & $\tau(V_L) = \tau_1(\vec{z}_1) \cdot \dots \cdot \tau_n(\vec{z}_n)$

where $\vec{z}_i$ is the projection of $\vec{v}_L$ onto the variables in $\vec{z}_i$

• "similar models": exact same signature
same variables, same range of values
& same allowed interventions

- only thing that's different is their structural equations
- they make slightly different predictions for the same event

• Distance function "dmax"

$$d_{\max}(M_1, M_2) = \max_{X \leftarrow X \in I, u \in R(u)} d_v(M_1(u, X \leftarrow x), M_2(u, X \leftarrow x))$$

- d_v = state metric, ruler used to measure the gap between the 2 results

Ex. M_1 predicts temp 20°C &
 M_2 predicts temp $22^\circ\text{C} \rightarrow$
 $d_v = 2$.

- $d_{\max} \rightarrow$ worst-case scenario, single largest difference

• M_1 is a $d_{\max} \cdot \alpha$ approximation of M_2 if $d_{\max}(M_1, M_2) \leq \alpha$.

- no matter what action we take or what the environment looks like, model's prediction will never be more than α away from the true/reference model

- In other words, if you have 2 algorithms (models) trying to predict the same thing $\rightarrow$ $d_{\max}$ tells you the maximum possible error you would ever encounter when switching between them

- Definition 4.1: Distance function $d_{\max}$ on similar probabilistic causal models

$$* d_{\max}(M_1, M_2) = \max_{X \leftarrow x \in I} \left(\sum_{u \in R(u)} \Pr(u) \cdot d_v(M_1(u, X \leftarrow x), M_2(u, X \leftarrow x)) \right)$$

① $\max_{X \leftarrow x \in I}$: worst-case intervention, we check every possible action that you can take (I) & find the one where the 2 models disagree the most

② $\sum_{u \in R(u)} \Pr(u) (\dots)$ $\rightarrow$ Expected value [Average]. Instead of looking for worst case context we multiply the error in each context by the probability of that context happening

③ d_v : error metric, measures actual physical distance between M_1 final predicted state & M_2 final predicted state.

In short $\rightarrow$ "If I perform the most problematic action possible, what is the average error I should expect"

across all likely environments?"

- Probability measure on $R(V)$:

$$\Pr(\vec{v}) = \{\vec{u} : M(\vec{u}) = \vec{v}\}$$

- Intervention $\vec{X} \leftarrow \vec{x}$ also induces a probability

$\Pr^{\vec{X} \leftarrow \vec{x}}$ on $R(V)$:

$$\Pr^{\vec{X} \leftarrow \vec{x}}(\vec{v}) : \Pr(\{\vec{u} : M(u, \vec{X} \leftarrow \vec{x}) = \vec{v}\})$$

• the 2 different probability distributions obtained by the 2 ways of lifting an intervention must be equal.

• In many cases, the whole point of abstraction is to be able to exclude rare low level boundary cases

• Intervention distribution $\Pr_I$:

- probab. distrib. that tells us

how likely a high level intervention $(\vec{Y} \leftarrow \vec{y})$ is to have been caused/triggered by a specific low level intervention $(X \leftarrow x)$

- Ex. Allows model to say: "when I click this high level button there are 10 different ways the neurons could fire to make it happen but way #1 is 90% more likely than others)

- SID between similar models M_1 & M_2 is the number of pairs (x, y) s.t there

exists an $x \in R(x)$ s.t

$$P_{r_y}^{M_1, x \leftarrow x} \neq P_{r_y}^{M_2, x \leftarrow x}$$

- Issues with SID:

- ① compares just the effect of interventions on single variables, whereas we allow arbitrary interventions
- ② we are interested in how far apart distrib. are, not just the fact that they are different
- ③ we need a definition that applies to models that aren't similar since that's what's needed for approximate abstraction